

# AI-Based Microscopy Image Analysis for Predicting Urinary Tract and Renal Diseases

Group ID: 25-26J-510



# Our Research Team



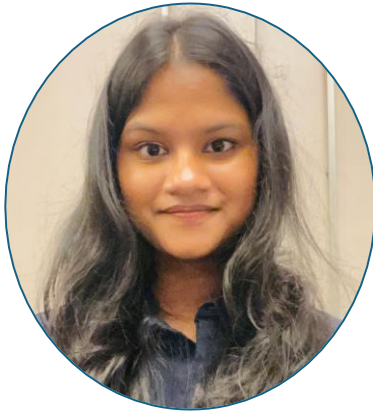
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# Introduction to the Overall Project

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- **Background**

- Urine microscopy is a widely used diagnostic method for urinary tract and renal diseases.
- **Sediment particles such as RBCs, WBCs, casts, crystals, and bacteria** are key biomarkers for the above conditions.
- Manual analysis is time-consuming, subjective, and prone to **inter-observer variability**.



# The Problem: Limitations of Current Diagnostic Systems

1

## Single Particle Focus

Most systems are designed to detect only a single type of particle, neglecting the intricate interplay between various biomarkers crucial for holistic diagnosis.

2

## Limited Metadata Integration

A significant drawback is the lack of integration with essential patient metadata, which reduces diagnostic accuracy and prevents personalized interpretations.

3

## Absence of Real-time Prediction

Crucially, existing solutions are not engineered for real-time, multi-disease prediction, slowing down urgent clinical decision-making processes.



# Research Gap



## Single Particle Focus

Current research predominantly concentrates on the detection of individual particle types, overlooking the synergistic information provided by multiple biomarkers.



## Lack of Disease-Level Prediction

Systems often stop at particle classification, failing to translate findings directly into comprehensive, interpretable disease-level predictions.



## Poor Patient Metadata Integration

There is a widespread absence of robust integration between microscopic image analysis and vital patient metadata, leading to less precise diagnoses.



## Lack Of Explainable AI (XAI) Support

The lack of Explainable AI (XAI) features limits clinical trust and understanding of AI-driven diagnostic recommendations, hindering adoption.



# Impact of Our Solution: Advancing Healthcare Diagnostics



## **Clinical Impact :**

Faster and more accurate diagnosis



## **Social Impact :**

Reduces diagnostic disparities in **resource-limited hospitals**



## **Research Impact**

Contributes to medical AI by developing a **multimodal diagnostic system**

# Previous Research vs. Our Innovative Approach

| Aspect               | Previous Research                             | Our Approach                                                          |
|----------------------|-----------------------------------------------|-----------------------------------------------------------------------|
| Focus                | Single particle detection                     | Multi-particle detection                                              |
| Disease Prediction   | Limited to particle-level classification only | Directly maps findings to diseases                                    |
| Metadata Integration | Rarely used                                   | Combines image features + patient metadata for personalized diagnosis |

# Constraints and Limitations



**Data Constraints:** Limited publicly available urine sediment datasets.



**Ethical Limitations:** Patient data and metadata require ethical clearance.

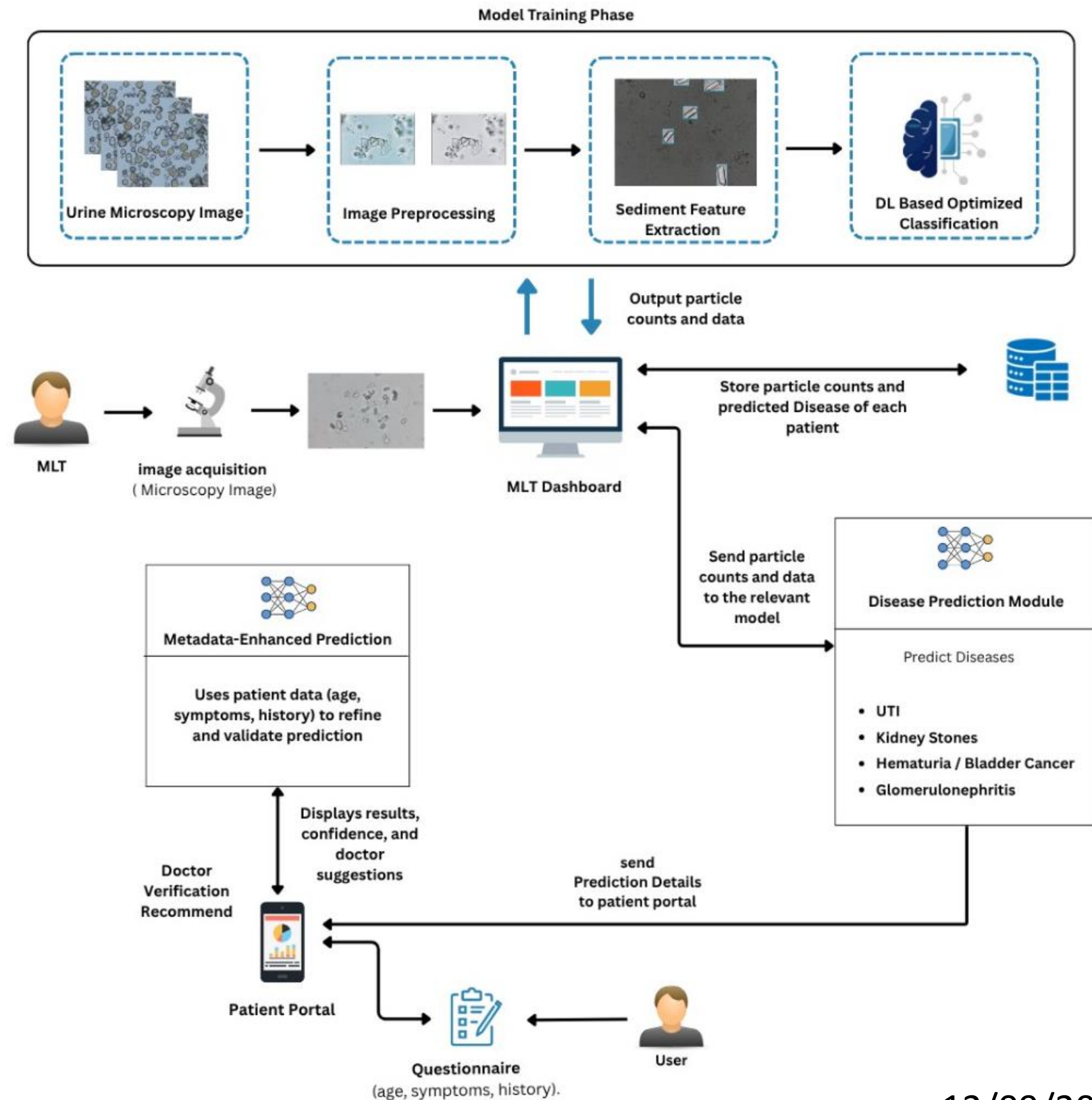


**Technical Constraints:** Requires high-quality microscopy images. High computational resources may be needed for real-time use.



**Clinical Constraints:** Model adoption depends on **clinician trust**.

# Overall System Diagram



# AI-based Identification Of WBCS and Bacteria In Urine Microscopy Images For UTI Classification

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**IT22221582 – Senanayake A.M.P.K.**

**Specialization: Software Engineering**

| Section                            | Summary Points                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proven Gap / Creative Solution     | <ul style="list-style-type: none"> <li>✓ Existing systems detect <b>only WBCs or only bacteria</b>.</li> <li>✓ Bacteria related to UTI are often treated as <b>one generic class</b>.</li> <li>✓ Current AI approaches mostly classified as <b>UTI vs Non-UTI</b> (binary).</li> </ul> <p><b>Creative Solution:</b> A <b>real-time hybrid AI pipeline</b> that:</p> <ul style="list-style-type: none"> <li>• Detects WBCs + bacteria simultaneously.</li> <li>• Classifies bacterial morphology (cocci, bacilli, spirochetes, chains/clusters).</li> <li>• Integrates WBC cast detection (team members module).</li> <li>• Produces <b>subtype classification</b> (Upper, Lower, No UTI).</li> </ul> |
| Knowledge Gap / Problem Definition | <ul style="list-style-type: none"> <li>• <b>Fragmented focus:</b> No system combining WBCs, bacteria, morphology, and casts.</li> <li>• <b>Manual microscopy:</b> Still time-consuming, subjective, error-prone.</li> <li>• <b>Metadata underutilized:</b> Current AI rarely integrates <b>patient</b> with image findings.</li> <li>• <b>UTI classification:</b> No system automates <b>UTI subtyping</b> from microscopy evidence.</li> </ul>                                                                                                                                                                                                                                                      |
| Shortcomings & Improvements        | <ul style="list-style-type: none"> <li>• Single-particle focus → <b>Multi-target detection of WBCs + bacteria</b></li> <li>• Bacteria treated as one class → <b>Morphology-based classification</b></li> <li>• Binary-only UTI classification → <b>Rule-based subtype</b></li> <li>• No metadata use → <b>Fuse microscopy findings + metadata for personalized diagnosis.</b></li> </ul>                                                                                                                                                                                                                                                                                                             |

# Specialization Knowledge

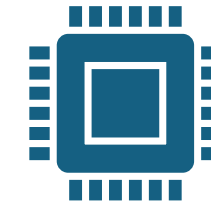


## Key Domains of Specialization

**Artificial Intelligence & Machine Learning** –  
CNNs, YOLO object detection.

**Computer Vision & Image Processing** –  
Preprocessing, segmentation, and  
morphology analysis.

**Medical Image Analysis** –  
Particle diagnostics + metadata integration.



## Latest Technologies Utilized

**Bacteria + WBC Detection** – YOLOv10,  
RT-DETR, EfficientDet-D7X

**Morphology classification** –  
EfficientNetV2, ConvNeXt, Swin  
Transformer

**OpenCV:** Image pre-processing

**PyTorch/Tensorflow:** building & training

**Labelling/CVAT:** To create datasets

# Specialization Knowledge

## Evaluation / Validation Strategy



**Precision, Recall, F1-score:** To ensure both correct detection and minimal false positives/negatives.

### Validation Methods

- **Cross-validation**
- **Train/Validation/Test Split**
- **External Validation**

### Clinical Relevance Validation

- Compare AI predictions with **manual lab technician diagnosis**.
- Correlate results with **urine culture reports** for real-world accuracy.

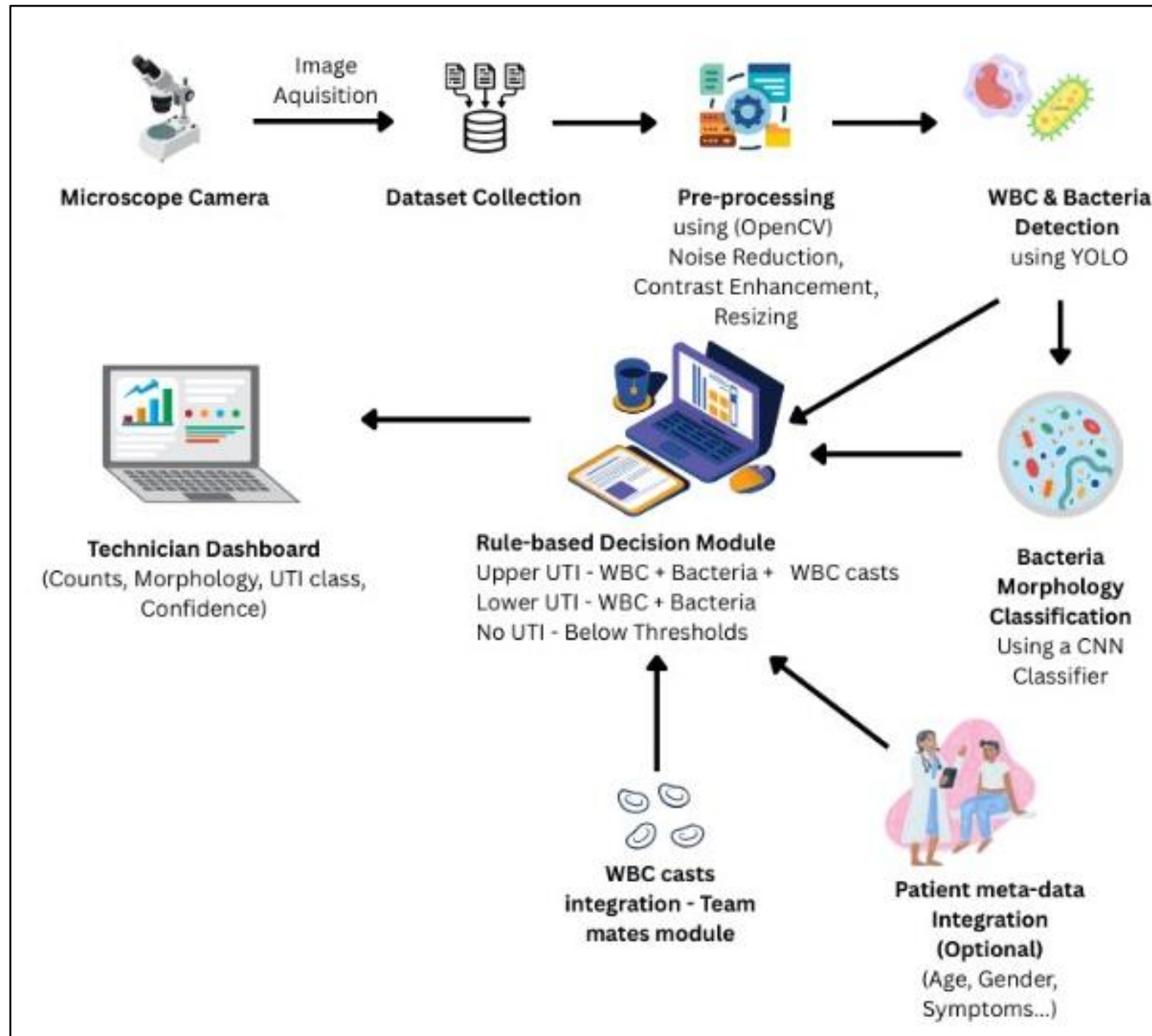
## Data Availability & Ethical Clearance



- Public datasets.
- Hospital-acquired microscopy images after ethical approval.
- Metadata collected with informed patient consent.

### Expected Outcomes:

- Detection of WBCs and bacteria across varied samples.
- Classification of bacterial morphology.
- UTI classification



## Real-world Usage

- Assists lab technicians by reducing manual workload & errors.
- Provides real-time standardized diagnosis in hospitals/clinics.
- Scalable to resource-limited settings with high patient flow.



# ▶ **AI-Based Detection and Classification of Urinary Crystals for Personalized Prediction of Kidney Stone Risk**

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**IT22589118 – Panagoda P.D.D.P.**

**Specialization: Software Engineering**

### Proven Gap / Creative Solution

- ✓ Current AI only detects crystals → no link to kidney stone risk.
- ✓ No integration of patient metadata (diet, hydration, family history, lifestyle).
- ✓ Proposed solution: **AI** Model for Identify Crystals and **personalized kidney stone risk prediction**.

### Knowledge Gap / Problem Definition

- ✓ Manual crystal analysis = **subjective, time-consuming, inconsistent**.
- ✓ Existing AI → focuses on **accuracy only**, ignores **clinical context**.
- ✓ No system connects **crystal morphology** → **disease prediction** → **prevention**.

### Shortcomings & Improvements

- ✓ **Few crystal types** → Detect all major + rare crystals.
- ✓ **No quantitative risk** → Counts/density per HPF + risk scores.
- ✓ **Ignore patient data** → Metadata fusion for personalized insights.
- ✓ **Offline/slow** → Real-time clinical deployment.
- ✓ **Small datasets** → Diverse, standardized datasets for generalizability.

# Specialization Knowledge

## Application of Specialization-Based Knowledge

- **Artificial Intelligence & Deep Learning** → for crystal detection and classification.
- **Computer Vision & Image Processing** → Preprocessing
- **Medical Informatics** → Integration of patient metadata (age, lifestyle, family history, hydration patterns) with crystal features.
- **Explainable AI (XAI)** → Transparency for clinical trust (visual reasoning, confidence scores).

## Latest Technologies Utilized

- **CNNs & YOLOv5** → Real-time crystal detection & classification.
- **Multimodal Fusion Models** → Combining image + metadata for personalized prediction.
- **React + Django/Flask** → Clinical dashboard, MLT interface, patient portal.

# Specialization Knowledge

## How to Evaluate / Prove Success

### ◆ Validation Methods

- **Train/Test Split & Cross-Validation** → ensure generalizability.
- **External Validation** → test with AI Prediction With MLT from multiple hospitals/Labs
- **Comparison with Manual Microscopy** → benchmark AI vs. expert lab technicians.
- Compare results with **urine culture reports** for real-world accuracy.

## Data Availability & Ethical Clearance

### ◆ Data Sources:

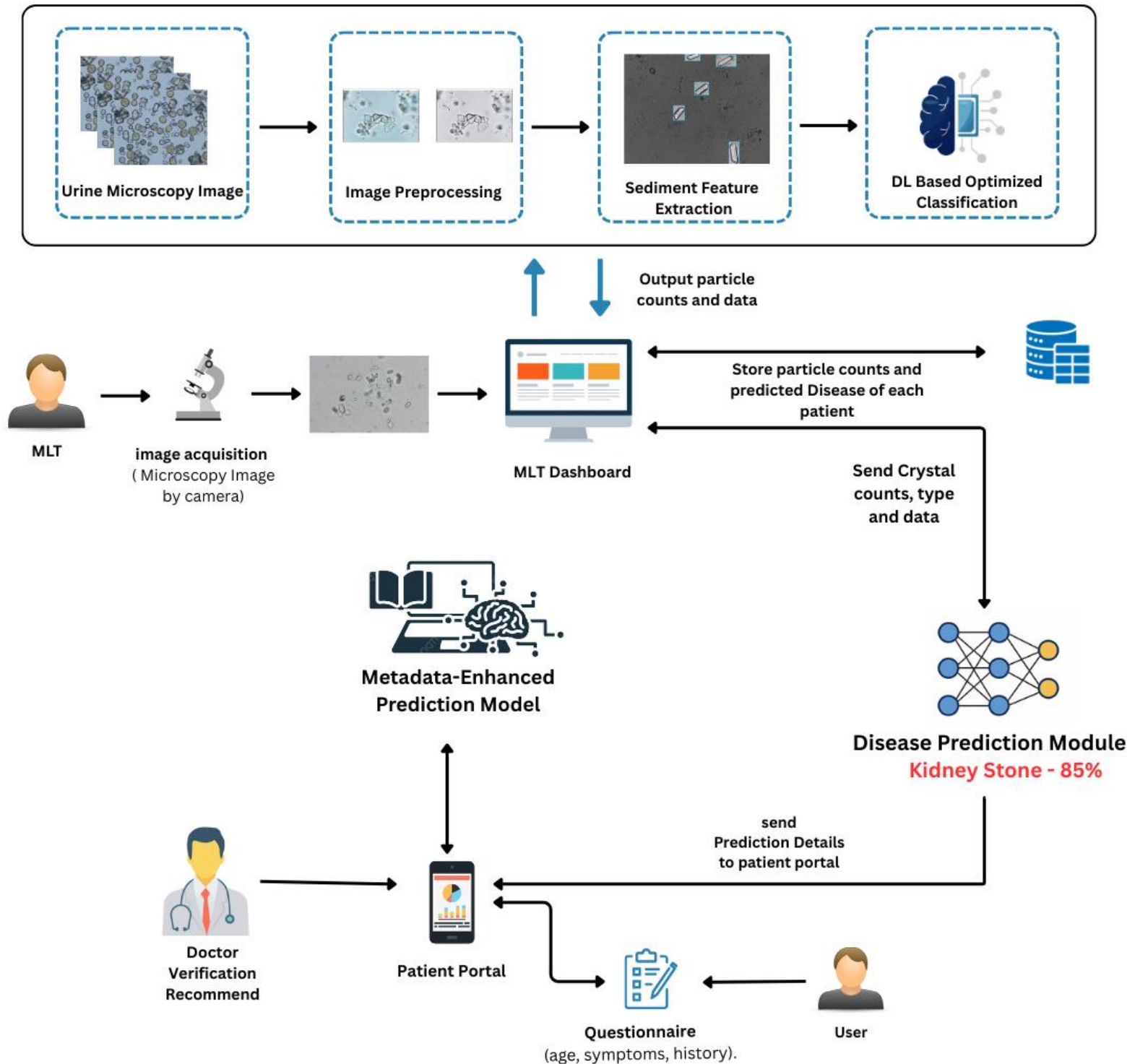
Public urinary crystal datasets (500–2000 images per type).

**New dataset collection from local hospitals** (microscopy images of calcium oxalate, uric acid, struvite, cystine).

# System Overview

## ◆ Real-World Scenario

- Patient visits clinic with **recurrent abdominal pain**.
- Urine microscopy images analyzed by AI within **2 seconds per frame**.
- AI detects **calcium oxalate monohydrate crystals**, integrates metadata (low water intake, family history).
- System generates **personalized risk score: 85% chance of kidney stone recurrence**.
- Clinician confirms output, prescribes preventive plan (hydration + citrate supplements).





# ▶ **AI-Based Detection and Classification of Urinary Casts for Kidney Disease Diagnosis**

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**Specialization: Software Engineering**

| Section                            | Summary Points                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proven Gap / Creative Solution     | <ul style="list-style-type: none"> <li>• Research still treats urinary casts as a single group, without linking them to specific kidney diseases.</li> <li>• No reliable automation exists for HPF (High Power Field) counting in urine microscopy.</li> <li>• <b>Proposed solution:</b> AI-based cast identification with automated counts, directly connected to disease prediction for better diagnosis.</li> </ul> |
| Knowledge Gap / Problem Definition | <ul style="list-style-type: none"> <li>• Manual Cast analysis – Miss subtype classification, time-consuming, inconsistent.</li> <li>• AI efforts focus on detection alone, with no linking results to diseases or defining HPF standards.</li> <li>• No proper validation or real-time use, which limits clinical impact.</li> </ul>                                                                                   |
| Shortcomings & Improvements        | <ul style="list-style-type: none"> <li>• <b>Treat casts as one class</b> → Subtype classification (RBC, WBC, granular, waxy).</li> <li>• <b>Manual HPF counts</b> → Automated HPF quantification.</li> <li>• <b>Ignoring patient data</b> – metadata fusion can provide personalized clinical insights.</li> </ul>                                                                                                     |

# Specialization Knowledge



## Key Domains of Specialization

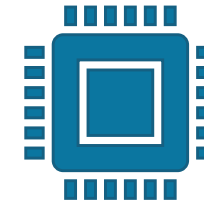
**AI & Deep Learning** → Detect and classify urinary casts.

**Image Processing** → Clean and normalize images for clear analysis.

**Quantitative Microscopy** → Count casts per HPF for accurate reporting.

**Medical Informatics** → Combine patient details with cast data for better diagnosis.

**Explainable AI** → Show heatmaps and confidence scores to build clinical trust.



## Latest Technologies Utilized

**YOLOv10 / RT-DETR** → Cutting edge models for real-time cast detection and classification.

**SAM 2 (Segment Anything)** → Advanced segmentation for accurate cast boundaries.

**ImageJ / Fiji Plugins** → Automated cast counting per HPF with precision.

**React + FastAPI (Python)** → Modern stack for interactive frontend and scalable backend.

# Specialization Knowledge



## Evaluation / Validation Strategy

**Train/Test Split & Cross-Validation** → Ensure robust performance of cast detection and classification.

**External Validation** → Test with microscopy images from multiple labs/hospitals to confirm generalizability.

**Comparison with Manual Microscopy** → Benchmark automated cast counts (per HPF) against expert lab technicians.

**Clinical Correlation** → Compare AI-predicted cast findings with patient diagnoses (e.g., glomerulonephritis, CKD) for real-world accuracy.



## Data Availability & Ethical Clearance

**Public urine sediment datasets** → Use as baseline for model training and testing.

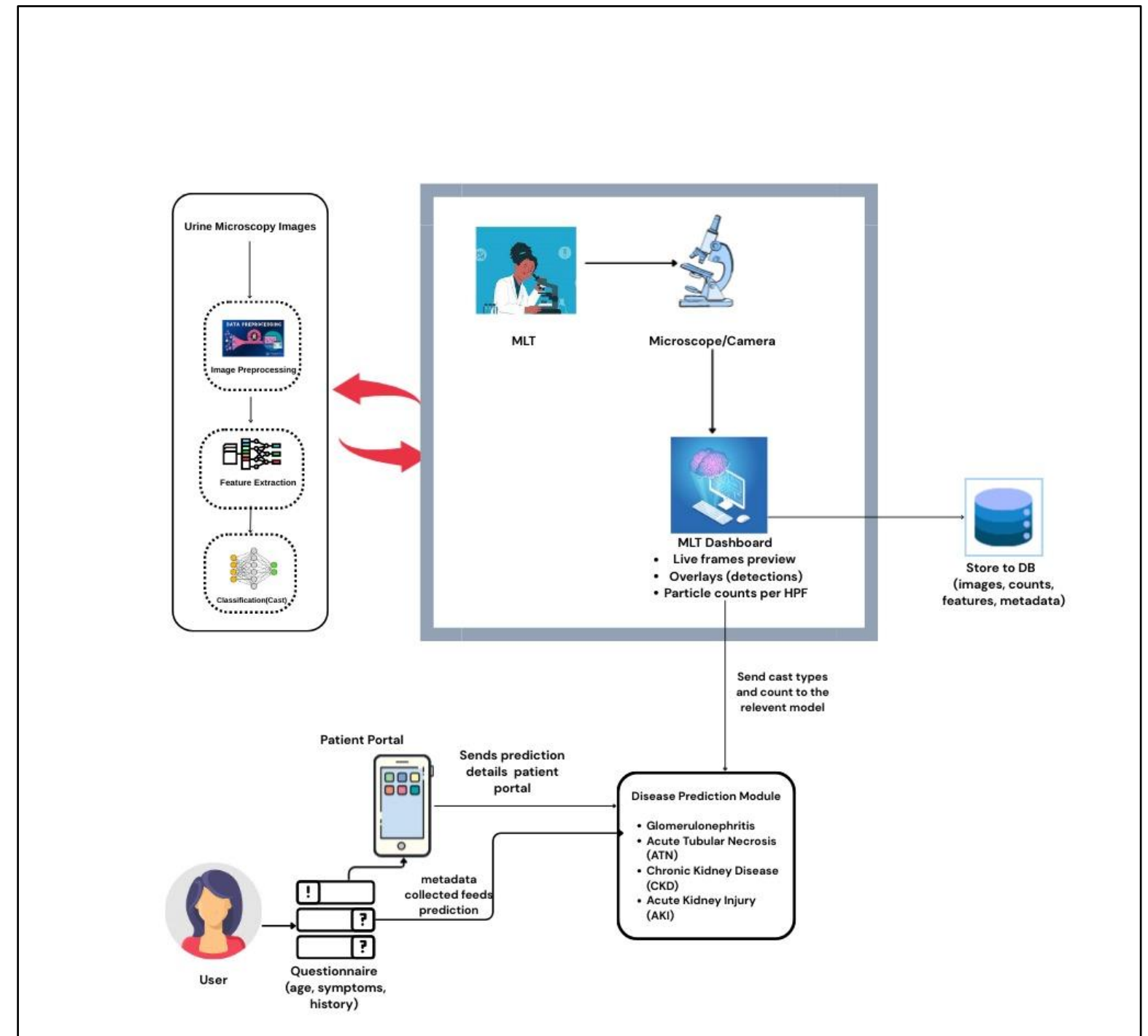
**Hospital microscopy images (with ethics approval)** → Collect real patient data for cast detection.

**Patient metadata (with informed consent)** → Age, lifestyle, history linked securely with image features.

# System Overview

## ◆ Real-World Scenario

- Patient arrives with symptoms of swelling and fatigue.
- Urine microscopy images are analyzed by AI within seconds.
- AI detects RBC and WBC casts, integrates patient metadata (age, hypertension history, hydration).'
- System flags high risk of glomerulonephritis, providing an explainable report with cast counts per HPF.
- Clinician reviews results, confirms diagnosis, and prescribes an early treatment plan.





# AI-based Detection Of RBCs In Urine Microscopy Images For Hematuria Cause Prediction

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**IT22590176 – Mathara Arachchi M.A.R.C.**

**Specialization: Software Engineering**

| Section                            | Summary Points                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proven Gap / Creative Solution     | <ul style="list-style-type: none"> <li>Existing AI detects only RBC counts, ignores morphology.</li> <li><b>Solution:</b> Deep learning (CNNs, YOLOv8) for iso vs dysmorphic classification</li> <li>Metadata fusion with patient history.</li> </ul>                                                                                                                                                                                                                                             |
| Knowledge Gap / Problem Definition | <ul style="list-style-type: none"> <li>Manual microscopy → subjective, inconsistent, and slow.</li> <li>No integration of patient data with imaging pipelines</li> <li>Limited datasets → poor accuracy &amp; weak generalization</li> </ul>                                                                                                                                                                                                                                                      |
| Shortcomings & Improvements        | <ul style="list-style-type: none"> <li><b>Existing systems:</b> <ul style="list-style-type: none"> <li>✓ Only count RBCs, ignore morphology</li> <li>✓ Small datasets, low reliability</li> <li>✓ No explainability for clinicians</li> </ul> </li> <li><b>Our improvements:</b> <ul style="list-style-type: none"> <li>✓ Morphology-aware AI (isomorphic vs dysmorphic)</li> <li>✓ Combine image + patient metadata</li> <li>✓ Explainable outputs (overlays, confidence)</li> </ul> </li> </ul> |

# Application of Specialization Knowledge

## Key Domains

- Machine Learning & Deep Learning
- Computer Vision (image preprocessing, feature extraction)
- Data Science (multimodal analysis)
- Software Engineering (deployment & security)

## Technologies

- PyTorch / TensorFlow
- YOLOv8, Faster R-CNN
- OpenCV / scikit-image
- Flask / FastAPI + React
- Docker / Kubernetes

## Evaluation

- Metrics: Accuracy, Sensitivity, Specificity, AUC
- Goal:  $\geq 90\%$  accuracy
- Validation: Real-world + synthetic datasets
- Cross-validation & external dataset testing for generalization

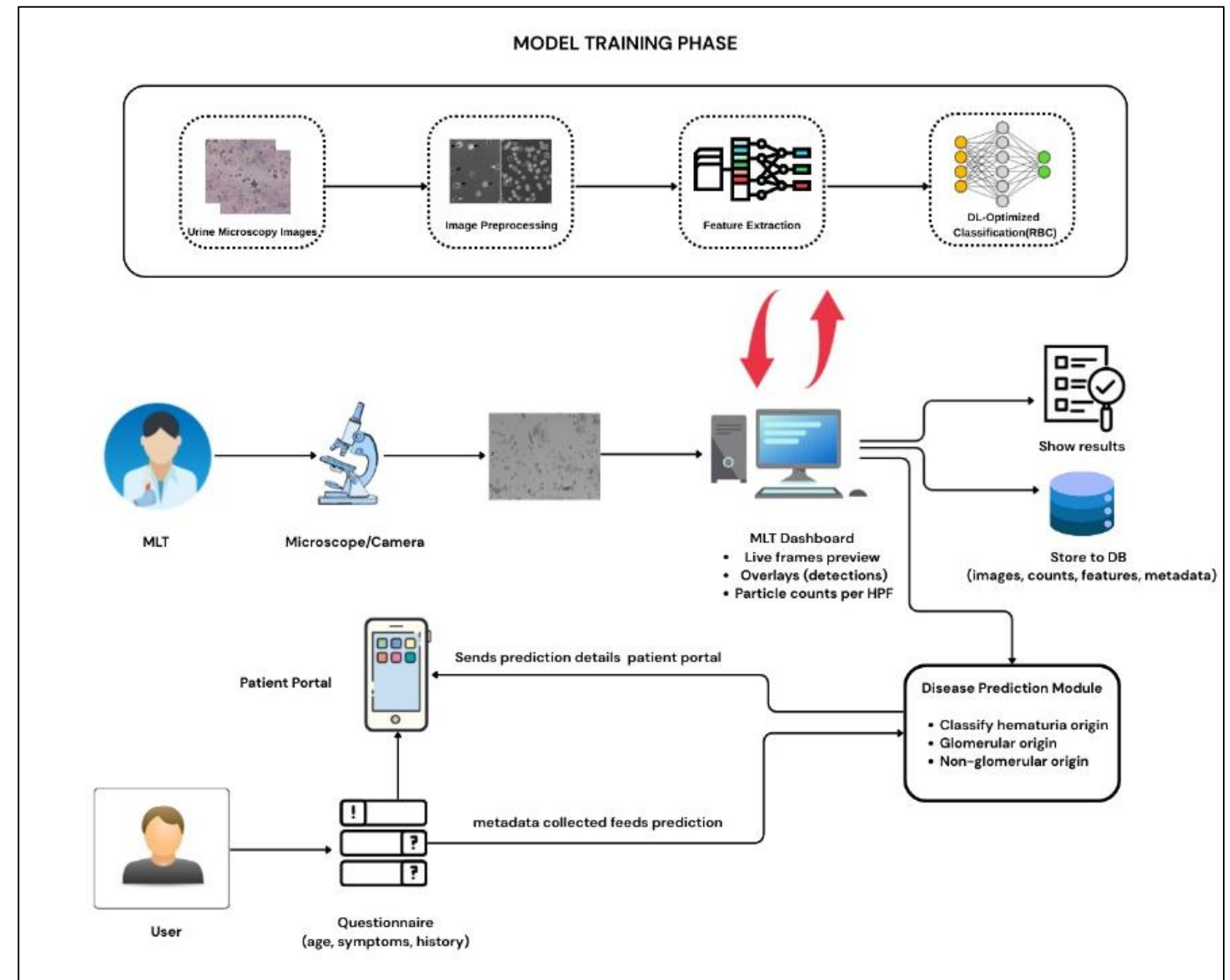
## Data & Ethics

- Hospital microscopy datasets
- Ethical approval & anonymized data
- Encrypted storage & role-based access

# System Overview

## Real-World Scenario

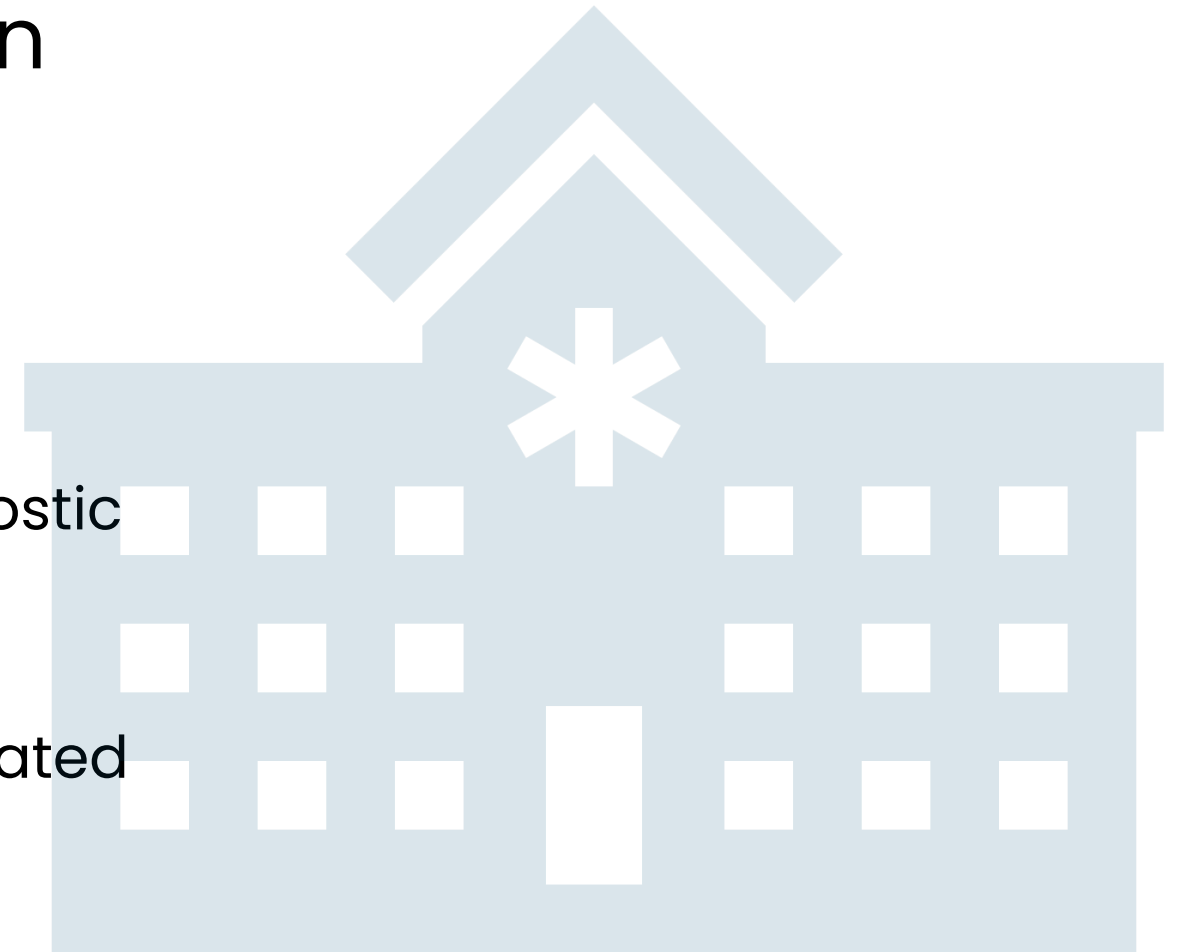
- Patient comes with **blood in urine**.
- **Microscopy image** is captured and uploaded.
- AI detects and classifies RBCs (isomorphic vs. dysmorphic).
- Metadata (age, history, habits) is added.
- System predicts **glomerular vs. non-glomerular hematuria**.
- **Doctor dashboard** → detailed results + overlays.
- **Patient portal** → simple report + next-step guidance.





## Ability of Commercialization

- High potential for adoption in hospitals, diagnostic labs, and telemedicine platforms.
- Can be packaged as a SaaS solution or integrated with hospital LIS (Lab Information Systems).



# Investment / Cost & Recovery

- **Initial Costs:**

- Labor: AI engineers, clinicians, annotators.
- Material: A camera for a microscope, GPUs, cloud services, storage, and microscopy datasets.

- **Cost Recovery:**

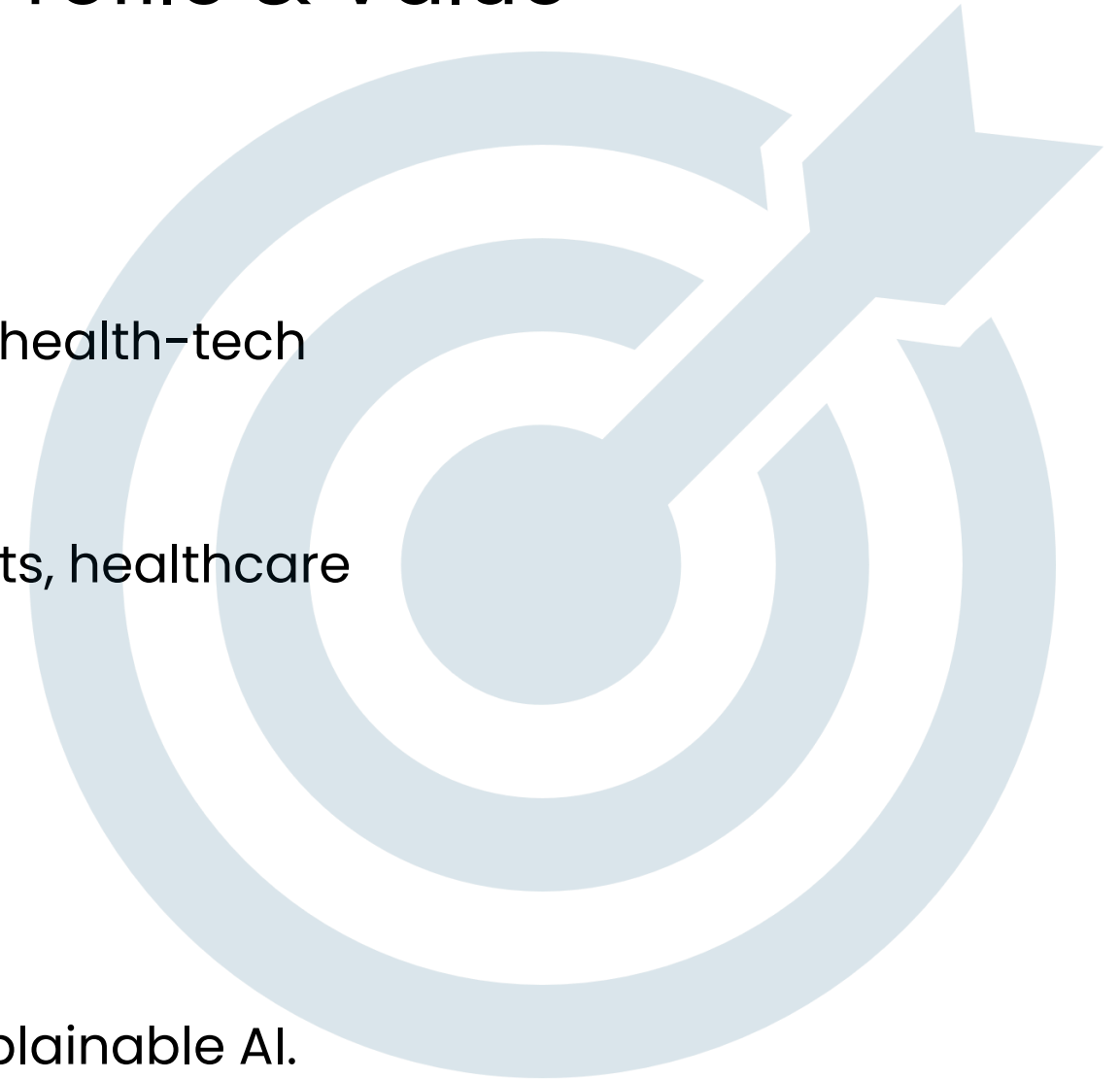
- Subscription-based licensing for labs/hospitals.
- Pay-per-use diagnostic API for telemedicine providers.





# Target Market, Customer Profile & Value Proposition

- **Target Market:** Hospitals, pathology labs, rural clinics, health-tech companies.
- **Customer Profile:** Laboratory technicians, nephrologists, healthcare providers.
- **Value Proposition:**
  - Automates manual microscopy → saves time.
  - Improves accuracy, consistency, and trust via explainable AI.
  - Reduces workload in resource-limited labs.





**Intellectual Property Rights**



Patentable **AI diagnostic framework** combining image + metadata analysis.



Proprietary annotated microscopy datasets.



Trademark for clinical web platform & dashboard.



# THANK YOU!